

Liam Moroy, PhD

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SUMMARY

AI Engineer / Research Associate with deep expertise in generative modelling (diffusion, score-based, LLMs) and end-to-end ML systems. Experienced across the full lifecycle: problem formulation, algorithm development, large-scale multi-GPU training, evaluation, and reproducible deployment. Track record of independently designing and shipping ML systems (RAG pipelines, generative models), publishing peer-reviewed work, mentoring students, and collaborating across multi-disciplinary teams.

EXPERIENCE

AI Engineer / Research Associate, Generative AI for Inverse Problems Dec. 2025 – Present
Heriot-Watt University *Edinburgh, Scotland*

- Built diffusion and flow-matching generative models for solving highly ill-posed inverse problems across multiple imaging modalities (nuclear Computed Tomography, non-conventional optics, medical imaging), trained on limited-data regimes on an HPC cluster.
- Improved the posterior-sampling ability of generative models (diffusion, flow matching), producing samples more faithful to the true Bayesian posterior and yielding better reconstruction quality and better-calibrated uncertainty estimates for downstream scientific decision-making.
- Accelerated sampling 2 to 10 times against state-of-the-art baselines while preserving quality.
- Deployed trained diffusion and flow-matching models as REST inference endpoints via FastAPI, exposing reconstruction and posterior sampling as a reproducible service usable by non-ML collaborators.
- Collaborated within a multi-disciplinary team of statisticians, physicists, and engineers on high-impact scientific and medical imaging applications.

AI Intern, Deep Learning for Tomographic Reconstruction of Compressible Flows Feb. 2022 – Aug. 2022
ONERA, the French Aerospace Lab *Paris, France*

- Designed and evaluated prior models for Bayesian CT reconstruction of compressible flows, enabling reconstruction from limited measurements where classical priors failed.

AI Intern, Deep Learning for Nebula Removal in Astronomical Images Jul. 2021 – Sep. 2021
Astrophysics Laboratory of Bordeaux *Bordeaux, France*

- Trained a regressor for nebula removal using transfer learning from synthetic data, handling noisy ground-truth labels unsuitable for direct supervision.

SELECTED PROJECTS

Generative Modelling of Implied Volatility Surfaces | *Diffusion, PyTorch*

- Score-based diffusion model (VP-SDE, attention U-Net) trained on ~3,500 daily SPY implied-volatility surfaces (2010–2023); no-arbitrage consistency emerges from data without explicit constraints.
- Frames daily surface calibration as posterior sampling under a diffusion prior; benchmarked ODE/SDE samplers (Euler, Heun, DPM-Solver).
- Quantitative Finance, Generative AI, Diffusion Models.* github.com/LiamMry/iv-surface-diffusion

Retrieval-Augmented Generation (RAG) for Scientific Literature | *RAG, LLMs, Docker*

- End-to-end, fully self-hosted RAG system for querying arXiv papers: PDF parsing (GROBID), Sentence-Transformer embeddings, ChromaDB vector search, and local LLM inference (Ollama).
- Designed a modular, incrementally re-runnable pipeline (ingestion, chunking, embedding, indexing, retrieval, generation) for reproducible, low-cost deployment with no external API dependencies.
- RAG, LLMs, Vector Search, Embeddings, Docker.* github.com/LiamMry/Scientific-RAG-system

EDUCATION

Université Paris-Saclay & ONERA, the French Aerospace Lab Paris, France
Ph.D. in Generative AI (Diffusion Models) for Inverse Problems in Imaging 2022 – 2025

- Generative AI (diffusion models) for solving inverse problems in imaging, applied to tomographic imaging.

ENSEIRB-MATMECA (Bordeaux INP) & University of Bordeaux Bordeaux, France
MSc Electrical Engineering + MSc Complex Systems (concurrent) 2020 – 2022

ENSEIRB-MATMECA (Bordeaux INP) Bordeaux, France
BSc Electronics Engineering 2017 – 2020

RESEARCH ACTIVITIES

Selected Publications

- “On the Posterior Gap in Plug&Play Diffusion Methods for Sparse-View CT”, IEEE JSTSP 2026 Special Issue on High-Dimensional Imaging, [L. Moroy](#), G. Bourmaud, F. Champagnat, J.-F. Giovannelli.
- “Evaluating the Posterior Sampling Ability of Plug&Play Diffusion Methods in Sparse-View CT”, ICASSP 2025, [L. Moroy](#), G. Bourmaud, F. Champagnat, J.-F. Giovannelli.

Selected Oral Presentations

- “Improving the Posterior Sampling of PnP Diffusion Models by Super-Calibration: application to Poisson inverse problem”, 7th SLSIP Workshop, May 2026, Krujë, Albania.
- “Evaluating the Posterior Sampling Ability of Plug&Play Diffusion Methods in Sparse-View CT”, GdR IASIS: Advances in Learning-Based Image Restoration, Dec 2024, Henri Poincaré Institute, Paris.

TEACHING & COMMUNICATION

University of Bordeaux

- Master ISC, MSc teaching assistant: Introduction to noise and random signals (2023/24 & 2024/25).

ENSEIRB-MATMECA Engineering School

- Engineering cycle, MSc teaching assistant: Generative modelling with diffusion models (2024/25).
- Engineering cycle, Freshman teaching assistant: Introduction to programming in C/C++ (2023/24 & 2024/25).

Interactive Scientific Outreach

- Interactive in-browser demos of generative modelling methods (annealed Langevin sampling, diffusion posterior sampling for inpainting), aimed at technical and non-specialist audiences.

TECHNICAL SKILLS

Programming: Python, C, C++

Generative Modeling: Score-Based Generative Modeling, Flow-Matching, Rectified Flow-Matching, Autoregressive Modeling, Normalizing Flows, Variational Auto-Encoders

LLMs & GenAI Systems: Retrieval-Augmented Generation (RAG), LLM Inference (Ollama), Embeddings, Vector Databases (ChromaDB), Sentence Transformers, Prompt Engineering, Docker, LoRA fine-tuning

textbfMLOps & Deployment: FastAPI, REST APIs, Model Serving, Docker, Git/GitHub, Linux/Bash

ML & Scientific Computing: CUDA, GPU-Accelerated Computing, PyTorch, Slurm, HPC Clusters

Research Methods & Domains: Generative Models, Plug-and-Play Methods, Bayesian Inference, Inverse Problems, Uncertainty Quantification, Statistical Data Analysis, Experimental Design, Signal Processing, Image Processing

Languages: French (Native), English (Fluent; TOEIC 950), Spanish (B1; Intermediate)